BizFlow prototype documentation

Overview

BizFlow helps small shops in Ghana track sales, stock and credit. This is a clickable prototype, not a real app yet. This document lists what the prototype does and the rules we agreed.

- **Who it is for:** a shop owner with a few helpers, selling in Ghana cedis (GH₵). The owner sees everything. Staff sell and do only what the owner allows.
- **Sample shop:** Adwoa Provisions, owned by Adwoa Darko, with staff Esi Asare and Kojo Appiah and products such as Milo Tin 400g, Indomie Chicken and Key Soap Bar.

What is not real yet. These parts show the steps but do nothing real: the camera and scanning, reading photos and files, printing, WhatsApp, Excel downloads, SMS codes, plan payments and syncing between devices. All prices, plan limits and sample numbers are placeholders.

Planned real stack(MERN). A React and Vite web app that installs on phones and PCs, works offline by storing data on the device, and syncs to a Node and Express server with MongoDB Atlas. Hosting on Netlify and Render, with Capacitor for app stores later. Paystack for payments, an SMS provider for codes, and Claude model for reading photos and documents.

Screens and devices

BizFlow has two layouts: a phone app with tabs at the bottom, and a PC and tablet workspace with a side menu. A PC / Tablet switch at the top of the workspace shows both.

Area	Phone app	PC and tablet workspace
Home	Home: today's sales, profit, cash, MoMo and credit, New sale button, Needs attention list, sales and profit trend	Dashboard: the same figures and alerts in cards
Sell	Sell tab, with a Checkout sheet	Sell page with the cart beside the products

Area	Phone app	PC and tablet workspace
Stock	Stock tab; products open in a sheet	Stock page with a product table and a details panel
Credit	Credit tab	Credit page
Reports	Reports tab	Reports page
History	Under More: Activity	History page
Staff	Under More: Staff, Approvals	Staff page
Security	Under More: Devices	Security page, with a count of waiting requests
Plans	Under More: Plans	Plans page

- **Phone sheets:** product details, customers, checkout, day report, add product, scanning and similar screens slide up from the bottom.
- **Workspace panels:** on PC, Stock and Sell show the list and the details side by side. On tablet, the details sit below the list.
- **Top bar:** the logo, the shop name, an Online or Offline badge and a Lock button.
- **Install:** staff who join by link are shown how to add BizFlow to their home screen.

Welcome, sign-up and sign-in

A new device starts on Welcome. A device already linked to a shop starts on Who is working?, where each person taps their name and enters their 4-digit PIN.

Welcome shows the logo, a 14-day free trial tag, the headline "Know your shop's numbers every day", a sample of the home screen and five feature cards. Buttons: Create an account, I already have an account, and Set up a print station.

Owner sign-up has three steps:

1. Details: shop name, owner name, mobile number, a password of at least 8 characters, and a tick to agree to the Terms of Use and Privacy Policy.
2. Code: a 6-digit code sent by SMS (123456 in the demo).
3. Finish: type of shop and a 4-digit owner PIN.

Terms of Use and Privacy Policy. Both open from the sign-up agree box and from the bottom of Welcome. They are plain-language drafts that a lawyer in Ghana should check. The business name, address and support contacts are still placeholders.

- **Terms of Use** cover: who can use BizFlow (18 and over), keeping PINs secret, the shop owning its records, the free trial, paying by MoMo, no automatic renewal, no refunds for unused time, what happens when a plan ends, how long records are kept, and Ghana law.
- **Privacy Policy:** what is kept (owner, staff, shop records, customers, security logs, payments, but never the MoMo PIN), why, who else sees it (Paystack, an SMS company, WhatsApp when chosen, Render and MongoDB Atlas, Claude model for imported photos), storage and deletion, security, rights under Ghana's Data Protection Act, 2012 (Act 843) with complaints to the Data Protection Commission, customers' information, under-18s, and changes.

A number that already has a shop is told to sign in instead.

Signing in

- On a trusted device, staff and owner use their 4-digit PIN. PINs work offline.
- On a new device, the owner signs in with mobile number and password. This needs internet.
- Five wrong tries lock that person out for 30 seconds.
- Lock in the top bar returns to Who is working?

Adding staff. The owner adds a staff member and gets a join link to send by WhatsApp or SMS. The link works once and expires after 24 hours. The PIN is never in the link; the owner tells it to them. The owner then approves the new phone.

Selling

A sale is built in a cart and paid by cash, MoMo, both, or credit. It saves even when offline.

Building the cart

- Search by name or scan a barcode. Each product shows its price and how many are left.
- Tap + to add one piece. Products sold in packs also have a pack button, such as "Carton of 6 · GH₵335.00", which adds a whole pack.
- A sale can mix packs and loose pieces, up to what is in stock. When no more fit, the button says why, such as "Only 2 pcs left".

- You can also type an amount in the number box.
- Products sold only by the pack show "Use the carton button" instead of +.
- Each person has their own cart.
- **Park sale** holds the cart so you can serve the next customer. You can pick it up again later or drop it.

Checkout

1. Choose Cash, MoMo, Cash + MoMo or Credit. For Cash + MoMo, type the cash part; the rest is MoMo.
2. To give a discount, type the amount in GH₵. The total drops and shows what it was before. The discount comes off profit too and shows on the receipt. Staff need the Give discounts permission.
3. For credit, choose the customer. Customers are added on the Credit page.
4. For MoMo, if the owner has set up Get paid by MoMo, type the customer's number and tap Send MoMo prompt. The customer approves with their PIN and the sale completes by itself. If they don't approve, try again or take cash. Without internet or without the setup, the customer sends to the shop's number and staff tap Complete sale.
5. Tap Complete sale, then print or send the receipt.

After the sale

- My sales today lists the signed-in person's own sales.
- Correct this sale removes items from a saved sale and asks why. Packs can only be removed whole.
- Corrections never add items. Record a new sale for anything extra.
- Staff corrections wait for the owner's approval, unless the staff member has "Correct sales without approval". Corrected sales are marked on the receipt, and reports use the new amounts.
- At the end of a shift, the cash count compares the cash expected with the cash counted.

Barcodes and scanning

Barcodes are optional and part of the first release. A product can have a piece barcode and a pack barcode. Anyone who can sell can scan.

The piece barcode box only shows when the product has a piece price. An unknown code can be added to a product.

Where	Phone and tablet	PC
Sell	A Scan button beside the search opens the camera. Each scan goes straight into the cart; scanning continues until Done	A USB barcode scanner types into the search box and the item goes into the cart. A note says "Have a barcode scanner? Just scan."
Stock	A Scan button beside the search opens the camera and then that product. Nothing goes into the cart	The USB scanner types the code and the list narrows to the product
Add product and Change barcodes	A Scan button beside each barcode box fills it from the camera	Type or use the USB scanner

- **Pack barcodes:** scanning a carton's barcode adds a whole pack at the pack price.
- **Unknown barcode:** the owner can add it as a new product. Staff are told to ask the owner.
- **Duplicates:** the same barcode can't be used for two products.
- **No Scan button on PC,** because webcams read barcodes poorly. Use a USB scanner.
- **In the prototype** the camera is not real, so you tap a sample barcode instead.

Stock

Stock is always counted in pieces. Packs are just a way to add and sell it. Every change goes into the product's stock history.

A product has: a name, how it is bought (carton, bag, pack, box, set or single pieces), pieces per pack, cost per piece, piece price, pack price, a low stock alert level and two optional barcodes.

Adding a product asks three things: how you buy it, how you sell it, and how many you have. The owner types what one carton cost, and the app works out the cost of one piece. It also shows the profit as prices are typed, in red if a price is below cost.

What one unit is called: stock is counted in pieces by default. A shop can call one unit something else, such as a tablet or sachet. The app then uses that word everywhere, for example "2 boxes, 37 tablets". The owner can change it later in the product's details. Only the name changes.

The choices are pieces, tablets, capsules, sachets, strips, bottles, pairs and tins. **Other…** lets the owner type their own name, such as roll.

Rounding: the cost of one piece is kept exact behind the scenes. Screens show amounts to 2 decimal places, and report totals always add up.

- **Pack-only products:** a wholesale shop can set only a pack price and leave the piece price blank. At least one price is needed.
- **Opening stock:** enter full packs and loose pieces. The app shows the total, for example 3 packs of 24 plus 6 = 78 pcs.
- **Adding stock later:** Restock asks how many packs came in. If the supplier's price changed, the owner can enter the new cost, and the app averages it with the old stock.
- **Undo a restock:** for 10 minutes, the person who added the stock or the owner can undo it. It can't be undone once some of that stock has been sold.
- **Removing stock:** needs a reason, such as damaged, expired or missing.
- **Low stock:** products at or below their alert level are tagged Low stock and listed first.
- **To buy list:** shows every low product and how many packs to buy. It can be sent on WhatsApp without prices. All staff can see it.
- **Stock count:** staff count the shelf without seeing the expected numbers. The owner then sees what is missing or extra and decides whether to update stock. Business plan and free trial only.
- **Cost price:** only the owner and staff allowed to see cost can see it. "Fix a wrong cost price" corrects a typing mistake and adds no stock.
- **Export to Excel:** Business plan and above.

Add from a photo or file (owner only)

- Upload a photo, PDF, Excel, CSV or Word file up to 10 MB, such as a price list or invoice.
- The app reads it into a list for the owner to check. If it can't tell how a product was bought, it marks the line CHECK and the owner chooses. For products already in stock, the owner chooses to add or skip.
- Each file uses one photo scan from the plan. Files added offline are read when back online.

- Download template gives an Excel sheet with the columns the app reads.
- Imported products need a price. Barcodes in the file are kept.
- In the prototype, any file gives the same sample list.

Credit customers

The Credit page shows who owes the shop and how much.

- **Adding a customer:** name and phone number. Two customers can't have the same name.
- **Selling on credit:** choose Credit at checkout, then the customer. The amount is added to their balance.
- **Recording a payment:** open the customer, type the amount and tap Record payment.
- **Reminders:** Send reminder opens a WhatsApp message to the customer (simulated).
- **Who can do it:** staff need "Sell on credit" to sell on credit and "Record credit payments" to record payments.
- **In reports:** a credit sale counts on the day it is made. Payments later are not counted again.

Deposits

The Credit page has two tabs: **Owing** and **Deposits**. A deposit lets a customer pay part now and collect the item later. The item is kept aside until it is fully paid.

- **Taking a deposit:** choose the customer and the item, then enter what they paid today.
- **Stock:** the item is set aside at once, so nobody else can buy it.
- **Paying in parts:** each payment is recorded. A bar shows how much is paid.
- **Collecting:** when it is fully paid, tap "Customer collected it". It then counts as a sale.
- **Cancelling:** owner only. The item goes back on sale, and the owner chooses whether to give the money back.
- **Cash:** deposit payments are included in the end-of-shift cash count.
- **Who can do it:** the same staff permissions as credit.

Suppliers

Owner only. A third tab, Suppliers, shows how much the shop owes the people it buys stock from. The owner adds a supplier with a name and phone number. For each one they can record "Bought on credit" or "I paid them", call or WhatsApp them, and see the last

few payments. A payment can't be more than what is owed.

Dashboard, reports and history

The owner sees today's numbers on Home, trends in Reports, and every action in a log that can't be changed.

Home and Dashboard

- Today's sales, number of sales and profit so far.
- Today by payment type: cash, MoMo and credit.
- Needs attention: requests waiting for approval, low stock, money owed and stock counts that came up short.
- Best sellers today.

Reports

- **Sales and profit trend:** a chart over 7 days, 30 days or 13 weeks. Tap a point to see that day.
- **Revenue by staff:** ranked by money taken (Business and above).
- **Day report:** the day's sales, profit, cash, MoMo, credit and stock problems, sent to WhatsApp (Business and above).
- **Missing stock:** what stock counts found short, who counted, and what it cost.
- Corrected sales always show their corrected amounts.

History and activity log

- Every sale, correction, stock change, price change, sign-in and approval is logged with who and when.
- Filter by person, date and type.
- Starter keeps the last 3 months; Business and above keep everything.

Staff, permissions and security

Every staff member can sell. Anything else is a permission the owner turns on for each person. New staff start with none.

Permission	What it allows
See cost and profit	Cost prices, profit per piece and profit totals
Change selling prices	Edit a product's selling price
Add stock	Add full packs to a product
Remove stock	Remove stock with a reason
Sell on credit	Choose Credit at checkout
Record credit payments	Record money paid by a credit customer
Correct sales without approval	Corrections apply at once instead of waiting for the owner
Give discounts	Take money off a sale at checkout. Off for new staff
Count stock	Do stock counts. Only on plans that include stock count
Send receipts by SMS	Send a receipt by SMS. On for new staff

Setting permissions needs the Business plan or above. On Starter, staff can only sell.

Managing staff

- Add staff with a name and phone number, then send them the join link.
- Removing a staff member stops their sign-in. Their past records stay.
- Staff limits: 2 on Starter and the trial, 10 on Business, unlimited on Multi-shop.

Devices and approvals

- The owner must approve each phone or computer before anyone uses it.
- A new device for the owner needs the phone number, password and a code. After that, PINs work on it, even offline.
- Device limits: 2 on Starter, 5 on Business, 5 per shop on Multi-shop.
- Approvals also list staff requests, such as sale corrections.

Locking

- The app locks itself after 30 seconds, 5 minutes (default) or 15 minutes of no use.
- Five wrong PINs lock that person out for 30 seconds.

My profile. Owner only. It has four parts, and every change goes into the activity log.

- **Your details:** name, shop name and type of shop. Works offline.
- **Mobile number:** a new number is confirmed with a code sent to it. Needs internet.
- **Password:** enter the old password, then a new one of at least 8 characters. Needs internet.
- **Your PIN:** enter the old PIN, then a new one. Easy PINs like 1234 are refused. Works offline.
- **Profile photo:** the owner can add a photo (JPG, PNG or WebP, up to 5 MB). Without one, their initials show instead. The photo also shows on the sign-in screen and in the top bar.
- **Get paid by MoMo:** the owner picks the network and enters their MoMo number. The app shows the registered name to check it, then asks for the Ghana Card number and an SMS code. Money is paid out the next working day. Changing the number also needs the SMS code, and the old number gets an SMS. Needs internet.
- **MoMo fee on the bill:** off by default. When the owner turns it on, a MoMo prompt adds about 1.95% as its own line, so the shop still gets the full price. The fee shows on the receipt and is not counted as sales or profit. Check with Paystack that this is allowed first.
- **Your data:** Download my data saves the shop, products, sales, customers, deposits, suppliers and payments in one file. PINs and passwords are never included. Delete my account asks the owner to type DELETE. The account closes and is deleted after 30 days; signing in before then keeps it.
- **Help and contact:** Contact us opens WhatsApp (+233 59 829 6141), call (+233 59 135 0125) and email (bizflowsupport.gh@gmail.com). It is also on the welcome page, the failed-payment box and the Terms and Privacy pages.

Receipts, printing and print station

Every sale has a receipt. It can be printed, sent to the customer, or sent from a staff phone to the shop computer to print.

What a receipt shows: shop name and phone, date, receipt number, who served, the items, the total and how it was paid. The owner sets these in Receipt details.

Ways to get it

- **Print:** uses the device's own print window.

- **Send to the customer:** by WhatsApp or SMS.
- **Print to the shop computer:** a staff phone sends the receipt to the print station. Both need internet. Receipts not printed in 30 minutes are dropped.

Setting up a print station

1. On the shop computer, open BizFlow and choose Set up a print station.
2. The owner shows a code on their own device. It works once, for 30 minutes.
3. Enter the code on the computer, then the owner approves it.

A shop can have more than one print station: 1 on Starter, 2 on Business and the free trial, 3 per shop on Multi-shop. With two or more, staff pick which station to print at, and the phone remembers it.

Export, WhatsApp and offline use

Selling works without internet on every plan. Records can be downloaded as Excel or shared on WhatsApp.

- **Export to Excel:** sales and stock. Owner only, Business plan and above.
- **WhatsApp:** the day report to the owner, credit reminders to customers, receipts to customers and staff join links.
- **SMS:** sign-up codes, receipts and staff join links.
- **Offline:** sales, stock changes and PIN sign-ins keep working, and sync when the internet is back.
- **Needs internet:** signing in on a new device, joining by link and reading photos or documents.

Plans and pricing

After a 14-day free trial there are three plans: Starter (GH₵49 a month), Business (GH₵99 a month) and Multi-shop (coming soon). Prices are samples.

Feature	Starter	Business	Multi-shop
Price a month	GH₵49	GH₵99	Soon
Price a year	GH₵490	GH₵990	Soon
Staff	2	10	Unlimited

Feature	Starter	Business	Multi-shop
Devices	2	5	5 per shop
History	3 months	All	All
Photo scans a month	20	200	500
Staff permissions	No	Yes	Yes
Revenue by staff	No	Yes	Yes
Expense tracking	No	No	Yes
Receipts	Yes	Yes	Yes
WhatsApp reports	No	Yes	Yes
Export to Excel	No	Yes	Yes
Stock count	No	Yes	Yes
MoMo prompts (Get paid by MoMo)	No	Yes	Yes
Several shops in one account	No	No	Yes

MoMo prompts also work during the free trial. If a shop moves to Starter, its saved payout number is kept and prompts come back when it upgrades.

- **Free trial:** 14 days with Business features, but only 2 staff and 20 photo scans.
- **Yearly billing:** 2 months free.
- **Selling never stops on any plan,** and plans can be changed at any time.
- **Locked features** say which plan has them instead of hiding.
- **Usage bars** on the Plans page show how much of each limit is used.
- **SMS:** each plan includes SMS each month: Trial 20, Starter 50, Business 200, Multi-shop 500. Receipts and join links sent by SMS use them. The owner can buy 100 more for GH₵10, and bought SMS never expire. WhatsApp is always free.

When a plan or free trial ends

A shop never loses its records and is never stopped in the middle of a sale.

1. **7 days before:** a reminder banner with a Pay with MoMo button, plus an SMS and WhatsApp reminder.
2. **7 days after:** everything still works.

3. **Limited mode:** after that, the owner can still sell, count stock, record credit and see the last 7 days. These stop until the owner pays:
    - Staff sign-in. Only the owner can sign in.
    - Print stations. Phones print on their own instead.
    - Photo scans, reports to WhatsApp, Excel export, revenue by staff, staff permissions and history older than 7 days.
4. **Paying again:** everything comes back exactly as before. Nothing is ever deleted.
5. **Take your records:** in limited mode the owner gets one free download of all records.

In the prototype, a row on the Plans page lets you jump in time to test these stages.

Paying with mobile money

All plan payments use one checkout. Only the owner sees it.

1. **Pay:** the owner picks 1 month or 1 year, then MTN MoMo, Telecel Cash or AirtelTigo Money, and checks the number.
2. **Approve on your phone:** the owner enters their MoMo PIN on their phone. The request runs out after 2 minutes.
3. **Result:**
    - Approved: the plan opens and a receipt number is shown and sent by SMS.
    - Declined, not enough money or no answer: nothing is charged, and the owner can try again.
4. **Payments:** the Plans page lists every payment.

Moving to a smaller plan: no payment. The smaller plan starts when the current one ends.

In the prototype, buttons choose the result. The real app will use Paystack through the BizFlow server, in test mode during development so no real money moves.

Brand

The app is called BizFlow : an open box inside an open hexagon of connected points, with "BizFlow" in Source Code Pro, the monospace font of the owner's original logo.

Definition: Hexagon represent a situation of bad luck and a closed box metaphorically signifies inaccessibility. The opened hex means freedom and the opened box mean accessible.

- **Logo in the app:** the full logo: the mark with text.

- **App icon:** the mark without text, white on a dark blue rounded square.

Product decisions and business rules

These were agreed during design. A real build should keep them.

- **Pack returns are whole packs only.** A customer who wants fewer pieces cancels the pack and buys singles.
- **Barcodes are optional.** A shop can work fully by searching names.
- **One barcode, one product.** A code already on a product can't be given to another one.
- **Scanning follows the stock.** Once a product has a barcode, anyone who can sell can scan it.
- **Scan button on phone and tablet only.** On a PC, use a USB scanner.
- **Browser printing first.** Direct printing to receipt printers comes later. Plans limit how many print stations a shop can add, but printing never stops.
- **Logo and icon.** The full logo is used inside the app. The app icon shows the mark only.

Not built yet and open questions

Simulated in the prototype

These show on screen but do nothing real yet: scanning, reading photos and files, printing, WhatsApp, Excel downloads and plan payments.

Not designed yet

- Expense tracking screens (planned for Multi-shop).
- **Expenses.** Record rent, transport, electricity and staff pay, and take them off profit so it shows real profit.
- **Multi-shop switching.** Add a second shop and switch between shops from one account.
- Making barcodes for products that have none. Not for now.
- **Shared barcode catalogue.** When a shop scans an unknown code, BizFlow suggests a name ("Is this Milo Tin 400g?"). Only the code, name and size are shared, never prices or sales.

    - **Filled before launch** from Open Food Facts, shelves scanned by the owner, pilot shops and possibly GS1 Ghana.
    - **Grows as shops use it.** A name is suggested once at least 2 shops agree on it.
    - **Wrong names:** shops can report a wrong name. Names rejected twice stop being suggested.
    - **Terms and Privacy** need a line about this sharing.
- **Shop-type features.** The shop type chosen at sign-up turns on a few extras for that kind of shop. Fields for features that are off are never shown.
    - **Expiry dates** (pharmacies, provisions, cosmetics): an expiry date per delivery and "expiring soon" alerts. Optional per product. Build this first.
    - **Sizes and colours** (Clothing shops only): one product with versions such as M/Blue and L/Red, each with its own stock. The stock list shows one row with the total. At checkout, staff pick the size and colour. Build this second.
    - **Measured amounts** (building materials, fabric, rice): selling by kg, metre or litre, such as 2.5 m of cloth.
    - **Serial numbers** (phone shops): an IMEI or serial number on the receipt for warranty.
    - **Ingredients** (chop bars): selling a meal uses up rice, oil and so on. A bigger change.

Open questions

- **Scanning on cheap phones.** Not yet tried on a low-cost Android phone.
- **Print station and device limits.** Decided: print stations have their own limit and don't count as devices.
- **Currency.** Prices and amounts use GH₵ only. Other countries would need their own currency. Going international (a Card option in the checkout, and a country and currency chosen at sign-up) is set aside for now.
- **Imported products.** Products brought in from a photo or file need a piece price or full pack price added. Barcodes are kept when the file has them, and the alert level is 10 when it has none.
- **Barcode lookup.** An online lookup that fills in product names from a barcode was suggested as an optional extra.